GEOMETRIC REDUCEDNESS AT A POINT MAY BE CHECKED AFTER AN ARBITRARY FIELD EXTENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-varieties/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup, of the native theorem declarations with mathematical macros, and of the project's own no-op definitions of the reference display environments; no macro was invented or redefined.

Editorial note. The same-chapter reference is located at its \label line inside the pinned chapter file; the two cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Geometric reducedness at a point, tested after finite purely inseparable extensions. Same chapter, Lemma labelled lemma-check-only-finite-inseparable-extensions (source line 463, Stacks tag 035Y); invoked at proof lines 520-521. Supplies that checking reducedness of the local ring after every finite purely inseparable extension is enough; the transport of that condition from k" back to x is the counted content.
- Descent of reducedness along a faithfully flat local ring map. Algebra, Lemma labelled algebra-lemma-descent-reduced (canonical Stacks tag 033F); invoked at proof lines 518-519. Supplies that reducedness of the bigger local ring implies reducedness of the smaller one.
- Existence of a common extension of two field extensions. Fields, Lemma labelled fields-lemma-common-extension-field (canonical Stacks tag 0H7K); invoked at proof lines 509-510. Supplies the field k"' dominating both k" and k', over which the two base changes may be compared.

Source reference. Original lemma 0H7K: sources/stacks-chow/tags.txt, source lines 21460–21460. Cross-chapter standard prerequisite invoked once at proof lines 509-510 for the existence of the common extension k"'; the faithful flatness and descent arguments are retained in the counted span.

Source reference. Original lemma 033F: sources/stacks-chow/tags.txt, source lines 4054–4054. Cross-chapter standard prerequisite invoked once at proof lines 518-519 for descent of reducedness along a faithfully flat local ring map; the identification of the ring map as faithfully flat is retained in the counted span.

Source reference. Original lemma 035Y: sources/stacks-varieties/pinned-original-source.tex, source lines 463–463. Same-chapter pointer invoked at proof lines 520-521 for the criterion that finite purely inseparable test extensions suffice; the transport of reducedness to each such extension is retained in the counted span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let k be a field. Let X be a scheme over k. Let k'/k be a field extension. Let $x \in X$ be a point, and let $x' \in X_{k'}$ be a point lying over x. The following are equivalent*

(1) *X is geometrically reduced at x,*
(2) *$X_{k'}$ is geometrically reduced at x'.*

In particular, X is geometrically reduced over k if and only if $X_{k'}$ is geometrically reduced over k'.

Proof. It is clear that (1) implies (2). Assume (2). Let k''/k be a finite purely inseparable field extension and let $x'' \in X_{k''}$ be a point lying over x (actually it is unique). Choose a common field extension k'''/k of k''/k and k'/k, see Fields, Lemma 0H7K. Choose a point $x''' \in X_{k'''}$ lying over both x' and x''. Consider the map of local rings

$$\mathcal{O}_{X_{k''},x''} \longrightarrow \mathcal{O}_{X_{k'''},x'''}.$$

This is a flat local ring homomorphism and hence faithfully flat. By (2) we see that the local ring on the right is reduced. Thus by Algebra, Lemma 033F we conclude that $\mathcal{O}_{X_{k''},x''}$ is reduced. Thus by Lemma 035Y we conclude that X is geometrically reduced at x. □